MA2508 Midterm Tutorial

Solution 1

By the chain rule,

$$\frac{\partial z}{\partial x} = 2xf'(x^2 - y^2), \qquad \frac{\partial z}{\partial y} = 1 - 2yf'(x^2 - y^2).$$

Therefore

$$\begin{aligned} y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} &= 2xyf'(x^2 - y^2) + x\big(1 - 2yf'(x^2 - y^2)\big) \\ &= 2xyf'(x^2 - y^2) + x - 2xyf'(x^2 - y^2) \\ &= \boxed{x}. \end{aligned}$$

Solution 2

- **True.**

  Method 1: Vector identity. In $\mathbb{R}^3$, the identity

  $$\|\vec{u} \times \vec{v}\|^2 + (\vec{u} \cdot \vec{v})^2 = \|\vec{u}\|^2\|\vec{v}\|^2$$

  gives $\|\vec{u}\|^2\|\vec{v}\|^2 = 0$. Hence at least one of the vectors has norm zero, so

  $$\boxed{\vec{u} = 0 \quad \text{or} \quad \vec{v} = 0.}$$

  Method 2: Geometric argument. Alternatively, if both were nonzero, a zero cross product would make them parallel, whereas a zero dot product would make them perpendicular. Both conditions cannot hold for two nonzero vectors. Therefore, at least one of the vectors must be zero.

  Method 3: Dot and cross product formulas. Suppose, for a contradiction, that $\vec{u} \neq \vec{0}$ and $\vec{v} \neq \vec{0}$. Then $|\vec{u}| > 0$ and $|\vec{v}| > 0$, and the angle $\theta \in [0, \pi]$ between the two vectors is defined.

  By the dot product formula,

  $$\vec{u} \cdot \vec{v} = |\vec{u}|\,|\vec{v}| \cos\theta = 0.$$

  Since $|\vec{u}|\,|\vec{v}| > 0$, we obtain $\cos\theta = 0$, hence $\theta = \pi/2$.

  The condition $\vec{u} \times \vec{v} = \vec{0}$ is equivalent to $|\vec{u} \times \vec{v}| = 0$. By the cross product magnitude formula,

  $$|\vec{u} \times \vec{v}| = |\vec{u}|\,|\vec{v}| \sin\theta = 0.$$

  Again dividing by the positive quantity $|\vec{u}|\,|\vec{v}|$ gives $\sin\theta = 0$, hence $\theta = 0$ or $\theta = \pi$.

  The same angle cannot equal $\pi/2$ and also belong to $\{0, \pi\}$. This is a contradiction. Therefore,

  $$\boxed{\vec{u} = \vec{0} \quad \text{or} \quad \vec{v} = \vec{0}.}$$

  Thus the statement is true.

- **False.** If f had continuous second-order partial derivatives, then it would require

  $$\boxed{f_{xy}(x, y) = f_{yx}(x, y)}$$

for every (x, y). However, the proposed first derivatives give

$$f_{xy}(x,y) = \frac{\partial}{\partial y}(f_x) = -2y, \qquad f_{yx}(x,y) = \frac{\partial}{\partial x}(f_y) = 1.$$

At $(0,0)$,

$$f_{xy}(0,0) = 0 \neq 1 = f_{yx}(0,0),$$

which contradicts $f_{xy} = f_{yx}$. Therefore such a function does not exist.

Solution 3

The derivatives of $\vec{r}(t)$ are

$$\vec{r}'(t) = e^t(\cos t - \sin t, \sin t + \cos t, 1),$$
$$\vec{r}''(t) = e^t(-2\sin t, 2\cos t, 1).$$

When $t = 0$, we have

$$\vec{r}'(0) = (1,1,1), \quad \vec{r}''(0) = (0,2,1).$$

Next, we compute cross product

$$\vec{r}'(0) \times \vec{r}''(0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 0 & 2 & 1 \end{vmatrix} = (-1,-1,2).$$

So, we have

$$|\vec{r}'(0) \times \vec{r}''(0)| = \sqrt{6}, \quad |\vec{r}'(0)| = \sqrt{3}.$$

and

$$\kappa(0) = \frac{|\vec{r}'(0) \times \vec{r}''(0)|}{|\vec{r}'(0)|^3} = \frac{\sqrt{6}}{3\sqrt{3}} = \boxed{\frac{\sqrt{2}}{3}}.$$

Solution 4

(a) The direction must be normalized:

$$\frac{\vec{u}}{\|\vec{u}\|} = \frac{(1,1)}{\sqrt{2}}, \qquad \nabla f = (-y\sin x, \cos x + 2y).$$

At $(0,1)$, $\nabla f = (0,3)$, so

$$\boxed{D_{\vec{u}} f(0,1) = \nabla f(0,1) \cdot \frac{\vec{u}}{\|\vec{u}\|} = \frac{3}{\sqrt{2}}.}$$

(b)

$$z - e^z + 2xy - 3 = 0.$$
$$F(x,y,z) = z - e^z + 2xy - 3$$
$$F_x = 2y, \quad F_y = 2x, \quad F_z = 1 - e^z.$$

at $(1,2,0)$, normal is $(4,2,0)$

$$\Rightarrow \text{ tangent plane } \quad 4(x-1) + 2(y-2) = 0 \quad \text{or} \quad 2x + y - 4 = 0.$$

(c) The normal vector of the ellipsoid is $(2x, 4y, 6z)$.

The tangent planes are parallel to $x + 3y + 5z = 7$, so we have

$$(2x, 4y, 6z) = k(1, 3, 5) \Rightarrow \begin{cases} x = \frac{k}{2} \\ y = \frac{3}{4}k \\ z = \frac{5}{6}k \end{cases}$$

Putting into the equation of ellipsoid, we have

$$\left(\frac{k}{2}\right)^2 + 2\left(\frac{3}{4}k\right)^2 + 3\left(\frac{5}{6}k\right)^2 = 498 \Rightarrow k^2 = 12^2 \Rightarrow k = \pm 12.$$

So, the tangent points are $\pm(6, 9, 10)$.

The tangent planes are

$$\begin{aligned} &(x-6) + 3(y-9) + 5(z-10) = 0 \\ \text{or} \quad &(x+6) + 3(y+9) + 5(z+10) = 0. \end{aligned}$$

Thus, we have

$$\boxed{x + 3y + 5z \pm 83 = 0}.$$

Solution 5

(a)

$$\begin{aligned} &\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{2 - e^{xy}} - 1} \\ &= \lim_{(x,y)\to(0,0)} \frac{xy(\sqrt{2 - e^{xy}} + 1)}{(\sqrt{2 - e^{xy}} - 1)(\sqrt{2 - e^{xy}} + 1)} \\ &= \lim_{(x,y)\to(0,0)} \frac{xy(\sqrt{2 - e^{xy}} + 1)}{1 - e^{xy}}, \quad \text{let } xy = a \\ &= \lim_{a\to 0} \frac{a}{1 - e^a} \cdot \lim_{a\to 0}(\sqrt{2 - e^a} + 1) \\ &= -1 \cdot 2 = -2. \end{aligned}$$

(b) Using polar coordinates $x = r\cos\theta$, $y = r\sin\theta$:

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} &= \lim_{r\to 0^+} \frac{r^3(\cos^3\theta - \sin^3\theta)}{r^2} \\ &= \lim_{r\to 0^+} r(\cos^3\theta - \sin^3\theta). \end{aligned}$$

Since $|\cos^3\theta - \sin^3\theta| \le 2$, we have:

$$0 \le |r(\cos^3\theta - \sin^3\theta)| \le 2r \to 0 \quad \text{as } r \to 0.$$

So, we have

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2} = 0}.$$

(c) Consider the limit on the line $y = mx$

$$\lim_{x\to 0} \frac{x(mx)^3}{x^2 + (mx)^6} = \lim_{x\to 0} \frac{m^3x^4}{x^2 + m^6x^6}$$
$$= \lim_{x\to 0} \frac{m^3x^2}{1 + m^6x^4} = \frac{m^3 \cdot 0}{1 + m^6 \cdot 0} = 0.$$

Consider the limit on the curve $x = y^3$

$$\lim_{y\to 0} \frac{y^3 \cdot y^3}{(y^3)^2 + y^6} = \lim_{y\to 0} \frac{y^6}{2y^6} = \frac{1}{2} \neq 0.$$

So,

$$\boxed{\lim_{(x,y)\to(0,0)} \frac{xy^3}{x^2 + y^6} \text{ does not exist.}}$$

Solution 6

(i) Because $x^2 + y^2 \geq 2|xy|$ for any $x, y \in \mathbb{R}$, then

$$|xy| \leq \frac{x^2 + y^2}{2}.$$

This means

$$0 \leq \left| \frac{xy}{\sqrt{x^2 + y^2}} \right| \leq \frac{\sqrt{x^2 + y^2}}{2}.$$

As $\lim_{(x,y)\to(0,0)} \sqrt{x^2 + y^2} = 0$, then

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2 + y^2}} = 0 = f(0,0).$$

So, $f(x, y)$ is continuous at $(0, 0)$.

Other proof

Let $x = r\cos\theta$, $y = r\sin\theta$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{r\to 0^+} \frac{r^2\cos\theta\sin\theta}{r} = \lim_{r\to 0^+} r\cos\theta\sin\theta = 0$$

for any value of θ.

So, $f(x, y)$ is continuous at $(0, 0)$.

(ii)

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h}$$
$$= \lim_{h\to 0} \frac{\frac{h\cdot 0}{\sqrt{h^2+0}} - 0}{h} = \boxed{0}.$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h) - f(0,0)}{h}$$
$$= \lim_{h\to 0} \frac{\frac{0\cdot h}{\sqrt{0+h^2}} - 0}{h} = \boxed{0}.$$

Solution 7

The first partial derivatives are:

$$\frac{\partial u}{\partial y} = z, \quad \frac{\partial u}{\partial z} = y, \quad \frac{\partial v}{\partial y} = xze^{xyz}, \quad \frac{\partial v}{\partial z} = xye^{xyz}.$$

Using the chain rule:

$$\frac{\partial w}{\partial y} = f_u \cdot \frac{\partial u}{\partial y} + f_v \cdot \frac{\partial v}{\partial y} = f_u \cdot z + f_v \cdot xze^{xyz}.$$

Differentiate $\dfrac{\partial w}{\partial y}$ with respect to z:

$$\begin{aligned}\frac{\partial^2 w}{\partial y \partial z} &= \frac{\partial}{\partial z}(zf_u + xze^{xyz} f_v) \\ &= f_u + z \cdot \frac{\partial}{\partial z}(f_u) + xe^{xyz} f_v \\ &\quad + xze^{xyz} \cdot \frac{\partial}{\partial z}(f_v) + x^2 yze^{xyz} f_v.\end{aligned}$$

Compute $\dfrac{\partial}{\partial z}(f_u)$ and $\dfrac{\partial}{\partial z}(f_v)$ using the chain rule:

$$\begin{aligned}\frac{\partial}{\partial z}(f_u) &= f_{uu} \cdot \frac{\partial u}{\partial z} + f_{uv} \cdot \frac{\partial v}{\partial z} \\ &= f_{uu} \cdot y + f_{uv} \cdot xye^{xyz}, \\ \frac{\partial}{\partial z}(f_v) &= f_{vu} \cdot \frac{\partial u}{\partial z} + f_{vv} \cdot \frac{\partial v}{\partial z} \\ &= f_{vu} \cdot y + f_{vv} \cdot xye^{xyz}.\end{aligned}$$

As f has continuous second-order partial derivatives, we have $f_{uv} = f_{vu}$. Substituting back:

$$\begin{aligned}\frac{\partial^2 w}{\partial y \partial z} &= f_u + z(yf_{uu} + xye^{xyz} f_{uv}) + xe^{xyz} f_v \\ &\quad + xze^{xyz}(yf_{vu} + xye^{xyz} f_{vv}) + x^2 yze^{xyz} f_v \\ &= f_u + xe^{xyz} f_v + x^2 yze^{xyz} f_v \\ &\quad + yzf_{uu} + 2xyze^{xyz} f_{uv} + x^2 yze^{2xyz} f_{vv}.\end{aligned}$$

So we have

$$\boxed{\frac{\partial w}{\partial y} = zf_u + xze^{xyz} f_v}$$

$$\boxed{\begin{aligned}\frac{\partial^2 w}{\partial y \partial z} &= f_u + xe^{xyz} f_v + x^2 yze^{xyz} f_v \\ &\quad + yzf_{uu} + 2xyze^{xyz} f_{uv} + x^2 yze^{2xyz} f_{vv}\end{aligned}}$$

Solution 8

- Along any real-valued path $y = mx^a$, $a < 6$, approaching $(0, 0)$:

  (i) If $m = 0$, then $y = 0$, so $f(x, y) = 0$.

  (ii) If $m \neq 0$ and $y \leq 0$, then $f(x, y) = 0$.

(iii) If $m \neq 0$ and $y > 0$, then

$$\frac{y}{x^6} = |m|\,|x|^{a-6} \longrightarrow +\infty \qquad \text{as } x \to 0,$$

Thus $y > x^6$ for sufficiently small $|x|$, so $f(x, y) = 0$.

For $m \neq 0$, approaching $(0, 0)$ requires $a > 0$. Hence, for every path in the question,

$$\boxed{\lim_{\substack{(x,y)\to(0,0)\\ y=mx^a}} f(x, y) = 0.}$$

- Along $y = x^6/2$, for $x \neq 0$,

$$0 < \frac{x^6}{2} < x^6 \quad \Rightarrow \quad f\left(x, \frac{x^6}{2}\right) = 1.$$

Therefore,

$$\lim_{x\to 0} f\left(x, \frac{x^6}{2}\right) = 1 \neq 0 = f(0, 0).$$

Thus

$$\boxed{f \text{ is discontinuous at } (0, 0).}$$

Solution 9

Define $F(x, y, z, \lambda) = xy + 2z - \lambda(x^2 + y^2 + z^2 - 11)$. Then $\nabla F = \vec{0}$ gives us

$$\begin{cases} y = 2\lambda x & (1) \\ x = 2\lambda y & (2) \\ 2 = 2\lambda z & (3) \\ x^2 + y^2 + z^2 = 11. & (4) \end{cases}$$

From (1) and (2): $x = 4\lambda^2 x \Rightarrow x(1 - 4\lambda^2) = 0$.

Case 1: $x = 0$

Then (1) gives $y = 0$. Constraint: $z^2 = 11 \Rightarrow z = \pm\sqrt{11}$.

Critical points: $(0, 0, \sqrt{11})$, $(0, 0, -\sqrt{11})$.

The corresponding values are $f = 2\sqrt{11}$, $f = -2\sqrt{11}$.

Case 2: $\lambda = \pm\frac{1}{2}$

From (3): $z = \pm 2$. From (1) and (2): $y = \pm x$.

When $\lambda = \frac{1}{2}$, $z = 2$, $y = x$, we have from (4): $2x^2 + 4 = 11 \Rightarrow x^2 = \frac{7}{2}$.

Critical points: $\left(\sqrt{7/2}, \sqrt{7/2}, 2\right)$, $\left(-\sqrt{7/2}, -\sqrt{7/2}, 2\right)$.

The corresponding values are $f = \frac{7}{2} + 4 = \frac{15}{2}$.

When $\lambda = -\frac{1}{2}$, $z = -2$, $y = -x$, we have from (4): $2x^2 + 4 = 11 \Rightarrow x^2 = \frac{7}{2}$.

Critical points: $\left(\sqrt{7/2}, -\sqrt{7/2}, -2\right)$, $\left(-\sqrt{7/2}, \sqrt{7/2}, -2\right)$.

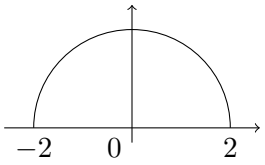
The corresponding values are $f = -\frac{7}{2} - 4 = -\frac{15}{2}$.

Comparing the above values, we have the following answer:

$$\boxed{f_{\max} = \frac{15}{2}}, \qquad \boxed{f_{\min} = -\frac{15}{2}}.$$

Solution 10

Max/min of $f(x,y) = x^2 + 2y^2 - x^2y^2$ in $D = \{x^2 + y^2 \le 4,\ y \ge 0\}$.



(1)
$$\begin{cases} f_x = 2x - 2xy^2 = 0 \\ f_y = 4y - 2x^2y = 0 \end{cases} \Rightarrow \begin{cases} x = \pm\sqrt{2} \\ y = 1 \end{cases} \qquad \text{and} \quad f(\pm\sqrt{2}, 1) = 2.$$

These 2 points are inside D.

(2) on $y = 0$, $f(x,0) = x^2$, $-2 \le x \le 2$

$$\Rightarrow f_{\min} = 0, \quad f_{\max} = 4. \quad \text{on } y = 0.$$

(3) on $x^2 + y^2 = 4$, $y > 0$.
let $L(x,y,\lambda) = x^2 + 2y^2 - x^2y^2 + \lambda(x^2 + y^2 - 4)$

$$\begin{cases} L_x = 2x - 2xy^2 + 2\lambda x = 0 \\ L_y = 4y - 2x^2y + 2\lambda y = 0 \\ L_\lambda = x^2 + y^2 - 4 = 0 \end{cases}$$

$$\begin{cases} 2x^2 - 2x^2y^2 + 2\lambda x^2 = 0 \\ 4y^2 - 2x^2y^2 + 2\lambda y^2 = 0 \\ x^2 + y^2 = 4 \end{cases} \Rightarrow \begin{cases} x = \pm\sqrt{5/2} \\ y = \sqrt{3/2} \end{cases} \quad \text{or} \quad \begin{cases} x = 0 \\ y = 2 \end{cases}$$

$$f(\pm\sqrt{5/2}, \sqrt{3/2}) = \frac{7}{4}, \qquad f(0,2) = 8.$$

(4) Compare all values $\Rightarrow$

$$\boxed{f_{\max} = 8, \qquad f_{\min} = 0}.$$

End of solutions